Neural Network Approximation of Elliptic PDEs via Residual Minimisation: Convergence, A Posteriori Bounds, and Error Decomposition

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Abstract

We develop a rigorous error analysis for approximating second-order elliptic partial differential equations using ReLU feedforward neural networks via dual-norm residual minimisation. For any coercive elliptic operator satisfying the Lax–Milgram conditions, we minimise the operator residual $\mathcal{R}(v) = \|Lv - f\|_{H^{-1}(\Omega)}$ over boundary-enforced ReLU classes $\mathcal{N}_n$, yielding H_0^1 error bounds from coercivity alone—without symmetry requirements or elliptic regularity.

We prove four main results with fully explicit constants: existence of global residual minimisers; quasi-optimal convergence at rate $O(n^{-1})$ for H^2 solutions matching the Céa constant M/α; a two-sided a posteriori bound; and a nonlinear projection characterisation extending Galerkin orthogonality to nonlinear trial manifolds.

The central contribution is a three-term error decomposition separating approximation, optimisation, and quadrature error with named constants for each term. The bounded-variation constant κ governing the quadrature term is proved finite for all $f \in \mathrm{BV} \cap L^\infty$ and all shallow ReLU networks, and is shown to grow as $O(n^{1/2})$ under weight-decay regularisation of strength $\Lambda = O(n^{-1/2})$. A joint parameter prescription in closed form guarantees a target accuracy η simultaneously across all three design parameters. All bounds are confirmed numerically across 1D, 2D, and non-symmetric convection-diffusion experiments, with independent verification of each error term.

Mathematics Subject Classification (2020). 35J05 (Primary); 65N15; 68T07; 46E35; 65N12; 41A46 (Secondary).

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Keywords: neural network approximation, elliptic PDEs, residual minimisation, quasi-optimality, a posteriori error bounds, bounded variation, three-term error decomposition.

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1 Introduction

The numerical approximation of elliptic partial differential equations is a central problem in scientific computing. Among elliptic problems, the Poisson equation

$$-\Delta u(x) = f(x), \quad x \in \Omega \subset \mathbb{R}^d, \quad u\big|_{\partial\Omega} = 0, \tag{1}$$

is the canonical model: it governs steady-state heat conduction, electrostatic potentials, incompressible pressure fields, and the Newtonian gravitational potential, and arises as the linearisation of many nonlinear elliptic systems. Its variational formulation in $H_0^1(\Omega)$ is the conceptual template on which finite element analysis [6, 8, 13] and neural-network-based PDE solvers [16, 21, 12] are built.

The classical variational formulation of (1) seeks $u \in H_0^1(\Omega)$ satisfying

$$a(u, v) := \int_\Omega \nabla u(x) \cdot \nabla v(x)\, dx = \langle f, v \rangle \quad \forall\, v \in H_0^1(\Omega). \tag{2}$$

The Lax–Milgram theorem [5, 13] guarantees a unique solution for any $f \in H^{-1}(\Omega)$. Classical Galerkin methods on finite-dimensional linear subspaces $V_n \subset H_0^1(\Omega)$ satisfy Céa's lemma [7]:

$$\|u - u_n\|_{H_0^1(\Omega)} \le \frac{M}{\alpha} \inf_{v \in V_n} \|u - v\|_{H_0^1(\Omega)}, \qquad M = \alpha = 1 \text{ for the Poisson equation.}$$

However, mesh-based methods suffer a fundamental computational obstacle in high dimensions: for target accuracy ε the required degrees of freedom scale as $N \sim \varepsilon^{-d}$ [3], making finite element methods prohibitively expensive when $d \ge 4$.

Neural networks offer a fundamentally different approximation paradigm. They form highly

adaptive, nonlinear function classes by composing affine maps with pointwise activations, and universal approximation results [9, 14] guarantee that any continuous function can be captured to arbitrary precision by a feedforward network. Quantitative Sobolev-norm approximation rates for deep ReLU networks have been established rigorously by Yarotsky [24] and Opschoor, Petersen, and Schwab [20], and crucially these rates can be made nearly dimension-independent for functions with compositional structure [20, 23]—a property that mesh-based methods cannot match.

The dual-norm residual approach. Despite rapid empirical progress, the theoretical error analysis of neural network PDE solvers remains incomplete. Physics-informed neural networks (PINNs) [21] minimise the squared L^2 strong residual, which does not directly yield $H_0^1(\Omega)$ error bounds without invoking elliptic regularity. The Deep Ritz method [12] minimises the energy functional, which requires symmetry of the bilinear form. Existing three-term error decompositions [10, 17, 19] provide important benchmarks; the present work advances this line by working in the H^{-1} dual norm, by providing named constants for each term—in particular an explicit bounded-variation constant κ for the quadrature term—and by deriving a joint balancing corollary that prescribes all three design parameters simultaneously in closed form.

This paper develops a complete error analysis for a dual-norm residual approach: we minimise the operator residual

$$\mathcal{R}(v) := \|Lv - f\|_{H^{-1}(\Omega)} = \| - \Delta v - f\|_{H^{-1}(\Omega)}$$

over boundary-enforced ReLU classes $\mathcal{N}_n$. The decisive analytical advantage is that coercivity of the bilinear form a directly converts the residual into an $H_0^1(\Omega)$ energy-norm error bound—without symmetry requirements or additional regularity assumptions beyond the Lax–Milgram conditions. The approach extends least-squares finite element methods [4] to nonlinear trial manifolds, and is closely related in spirit to the Weak Adversarial Network framework [25], from which it is distinguished by working with the exact H^{-1} dual norm rather than an adversarial approximation.

1.1 Main Contributions

We develop a self-contained, rigorous error analysis framework for dual-norm residual minimisation applied to elliptic PDEs. The specific advances relative to existing work are described below; comparisons with De Ryck and Mishra [10] and Liao and Ming [17] are made explicit where relevant.

(C1) Functional-analytic setup. Section 2 identifies all operator constants ($M = \alpha = 1$ for the Poisson equation), verifies the Lax–Milgram hypotheses in full, and gives a precise definition of the boundary-enforced ReLU class $\mathcal{N}_n = \{x(1-x)\Phi_\theta : \theta \in \mathbb{R}^n\}$ together with a density theorem and proof of density in $H_0^1(0,1)$ (Theorem 2.15).

(C2) Existence of global minimisers. Lemma 3.6 establishes existence of global residual minimisers in $\mathcal{N}_n$ via a compactness argument that handles the over-parameterisation difficulty through an explicit normalisation and δ-separation condition on the parameter set.

(C3) Quasi-optimal convergence with explicit rate. Theorem 4.1 proves $\|u - u_n\|_{H_0^1(\Omega)} \leq (M/\alpha) \inf_{v \in \mathcal{N}_n} \|u - v\|_{H_0^1(\Omega)}$ for any coercive operator satisfying Lax–Milgram, matching the Céa constant for classical Galerkin approximations on linear subspaces. For H^2 solutions, Corollary 4.2 gives the explicit rate $O(n^{-1})$.

(C4) Computable a posteriori error certificate. Lemma 4.3 provides a two-sided bound equating the exact H^{-1} residual to the H_0^1 error, with a detailed analysis of the computable L^2 residual proxy $\mathcal{R}_h$ via the Poincaré constant $C_P = 1/\pi$.

(C5) Three-term error decomposition with jointly optimal parameter balancing. Theorem 4.11 provides, for an ε-approximate minimiser of the discretised loss computed on N_q quadrature points, the bound

$$\|u - u_n^\varepsilon\|_{H_0^1(\Omega)} \leq \underbrace{\frac{M}{\alpha} \inf_{v \in \mathcal{N}_n} \|u - v\|_{H_0^1(\Omega)}}_{\text{(I) approximation}} + \underbrace{\frac{C_P}{\alpha}\varepsilon}_{\text{(II) optimisation}} + \underbrace{\frac{2C_P\kappa^{1/2}}{\alpha N_q^{1/2}}}_{\text{(III) quadrature}},$$

where every constant is explicitly identified. Relative to [10, 17], two specific advances are made: (a) the quadrature error is bounded explicitly via a bounded-variation argument with named constant κ, whose dependence on network width is characterised in Proposition 4.7 and verified numerically; and (b) Corollary 4.12 derives a jointly optimal prescription for all three design parameters simultaneously in closed form.

(C6) Nonlinear projection characterisation. Proposition 4.4 records that residual minimisers over $\mathcal{N}_n$ satisfy a generalised Galerkin orthogonality condition. This observation clarifies the geometric structure of the problem but does not constitute a new result independent of (C3).

(C7) Numerical validation. Section 6 presents experiments confirming the $O(n^{-1})$ rate and, additionally, independently varying ε and N_q to verify the optimisation and quadra-

ture terms of Theorem 4.11, and confirming the joint balancing prescription of Corollary 4.12.

1.2 Relation to Prior Work

Approximation theory for ReLU networks. Cybenko [9] and Hornik [14] established universal approximation; Barron [1] obtained dimension-independent rates in a spectral norm ball. Yarotsky [24] proved the first sharp $W^{1,\infty}$ rates for deep ReLU networks. Opschoor, Petersen, and Schwab [20] extended these to high-order Sobolev norms. Siegel and Xu [23] established tight H^1 approximation bounds for shallow ReLU networks.

Neural network methods for elliptic PDEs. The Deep Ritz method [12] minimises the energy functional; Müller and Zeinhofer [19] analysed its convergence and provided a three-term decomposition for that setting. PINNs [21] minimise the L^2 strong residual; Shin, Darbon, and Karniadakis [22] proved convergence for linear elliptic PDEs; Mishra and Molinaro [18] provided generalisation error bounds. De Ryck and Mishra [10] decomposed PINN error into approximation, optimisation, and statistical components for Kolmogorov PDEs. Liao and Ming [17] obtained an approximation–optimisation error decomposition for Deep Ritz. Doumpos, Eigel, and Schneider [11] developed residual-based error correction in direct dialogue with our a posteriori bound.

Positioning of Theorem 4.11 relative to [10, 17]. De Ryck and Mishra [10] decompose PINN error for Kolmogorov PDEs; their setting uses the L^2 strong residual, requiring elliptic regularity to control the H_0^1 error. Liao and Ming [17] provide a two-term approximation–optimisation decomposition for Deep Ritz without an explicit quadrature term. Müller and Zeinhofer [19] treat the Deep Ritz setting with symmetry requirements. The present work differs in three respects: (a) coercivity in the H^{-1} dual norm controls the H_0^1 energy-norm error directly, eliminating the need for elliptic regularity; (b) the quadrature error is bounded explicitly with named constant κ rather than absorbed into a discretisation term; (c) Corollary 4.12 prescribes all three design parameters jointly and simultaneously. The three-term decomposition structure itself is shared with prior work; these are the specific refinements within the dual-norm residual framework.

1.3 Paper Organisation

Section 2 develops the functional-analytic prerequisites. Section 3 defines the residual functional and proves existence of minimisers. Section 4 presents the main theoretical results.

Section 5 works through the symbolic finite-dimensional example. Section 6 presents numerical experiments. Section 7 provides a unified discussion. Section 8 discusses limitations and extensions. Section 9 concludes.

2 Functional-Analytic Preliminaries

Throughout, $\Omega \subset \mathbb{R}^d$ $(d \geq 1)$ denotes a bounded domain with Lipschitz boundary $\partial\Omega$. We write $\langle \cdot, \cdot \rangle$ for the duality pairing between $H^{-1}(\Omega)$ and $H_0^1(\Omega)$.

Notation. $L^2(\Omega)$ denotes the standard Lebesgue space with norm $\|\cdot\|_{L^2(\Omega)}$. $H^1(\Omega)$ is the first-order Sobolev space with norm $\|u\|_{H^1}^2 = \|u\|_{L^2}^2 + \|\nabla u\|_{L^2}^2$. $H_0^1(\Omega)$ is the subspace of $H^1(\Omega)$ with zero boundary trace, equipped with the seminorm $\|u\|_{H_0^1(\Omega)} = \|\nabla u\|_{L^2(\Omega)}$ (a full norm by the Poincaré inequality, Theorem 2.5). $H^{-1}(\Omega) = (H_0^1(\Omega))^*$ is the dual space with norm $\|F\|_{H^{-1}(\Omega)} = \sup\{|\langle F, v\rangle| : v \in H_0^1(\Omega), \|v\|_{H_0^1(\Omega)} \leq 1\}$.

2.1 Lebesgue and Sobolev Spaces

Definition 2.1 ($L^2(\Omega)$)**.** The Lebesgue space $L^2(\Omega)$ is the collection of (equivalence classes of) measurable functions $u : \Omega \to \mathbb{R}$ satisfying $\int_\Omega |u(x)|^2\, dx < \infty$, equipped with the inner product $(u, v)_{L^2} := \int_\Omega u(x)v(x)\, dx$ and norm $\|u\|_{L^2(\Omega)} := (\int_\Omega u^2\, dx)^{1/2}$.

Definition 2.2 (Weak partial derivative)**.** For $u \in L^2(\Omega)$ and multi-index α, the function $g \in L^2(\Omega)$ is the α-th weak partial derivative of u, written $D^\alpha u = g$, if $\int_\Omega u(x) D^\alpha \phi(x)\, dx = (-1)^{|\alpha|} \int_\Omega g(x)\phi(x)\, dx$ for all $\phi \in C_c^\infty(\Omega)$.

Definition 2.3 (Sobolev space $H^1(\Omega)$)**.** $H^1(\Omega) := \{u \in L^2(\Omega) : D^\alpha u \in L^2(\Omega)\ \forall\, |\alpha| \leq 1\}$, with inner product $\langle u, v\rangle_{H^1} = \int_\Omega uv\, dx + \sum_{i=1}^d \int_\Omega (\partial_i u)(\partial_i v)\, dx$.

Definition 2.4 ($H_0^1(\Omega)$)**.** $H_0^1(\Omega) \subset H^1(\Omega)$ is the closure of $C_c^\infty(\Omega)$ in the H^1-norm. For $\Omega = (0, 1)$: $H_0^1(0, 1) = \{u \in H^1(0, 1) : u(0) = u(1) = 0\}$.

Theorem 2.5 (Poincaré inequality [13, Thm. 5.8.1])**.** *Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain. There exists $C_P = C_P(\Omega) > 0$ such that $\|u\|_{L^2(\Omega)} \leq C_P \|\nabla u\|_{L^2(\Omega)}$ for all $u \in H_0^1(\Omega)$. For $\Omega = (0, 1)$ the optimal constant is $C_P = 1/\pi$.*

Proof. Let $u \in H_0^1(0, 1)$ and expand in the Fourier sine series: $u(x) = \sum_{k=1}^\infty a_k \sin(k\pi x)$, $u'(x) = \sum_{k=1}^\infty k\pi a_k \cos(k\pi x)$. By Parseval's theorem, $\|u\|_{L^2}^2 = \frac{1}{2}\sum_{k=1}^\infty a_k^2$ and $\|u'\|_{L^2}^2 = \frac{\pi^2}{2}\sum_{k=1}^\infty k^2 a_k^2$. Since $k^2 \geq 1$ for all $k \geq 1$, $\|u\|_{L^2}^2 \leq \frac{1}{\pi^2}\|u'\|_{L^2}^2$, so $\|u\|_{L^2} \leq \frac{1}{\pi}\|u'\|_{L^2}$. Equality is attained by $u_1(x) = \sin(\pi x)$, confirming $C_P = 1/\pi$ is sharp. $\square$

Definition 2.6 (Dual space $H^{-1}(\Omega)$)**.** $H^{-1}(\Omega) = (H_0^1(\Omega))^*$ consists of all bounded linear functionals $F : H_0^1(\Omega) \to \mathbb{R}$, with dual norm $\|F\|_{H^{-1}(\Omega)} := \sup\{|\langle F, v\rangle| : v \in H_0^1(\Omega), \|v\|_{H_0^1(\Omega)} \leq 1\}$. For $f \in L^2(\Omega)$, the functional $F_f(v) = \int_\Omega f v \, dx$ satisfies $\|F_f\|_{H^{-1}(\Omega)} \leq C_P \|f\|_{L^2(\Omega)}$.

2.2 Variational Formulation of the Poisson Equation

Definition 2.7 (Poisson boundary value problem)**.** Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and $f \in H^{-1}(\Omega)$. Find $u \in H_0^1(\Omega)$ such that

$$a(u, v) := \int_\Omega \nabla u(x) \cdot \nabla v(x) \, dx = \langle f, v\rangle \quad \forall\, v \in H_0^1(\Omega). \tag{3}$$

Proposition 2.8 (Properties of the Poisson bilinear form)**.** *The bilinear form* $a(u, v) = \int_\Omega \nabla u \cdot \nabla v \, dx$ *satisfies:*

(i) Symmetry: $a(u, v) = a(v, u)$.

(ii) Boundedness with $M = 1$: $|a(u, v)| \leq \|u\|_{H_0^1} \|v\|_{H_0^1}$.

(iii) Coercivity with $\alpha = 1$: $a(v, v) = \|v\|_{H_0^1}^2$.

In particular, $M/\alpha = 1$.

2.3 The Lax–Milgram Theorem

Assumption 2.9 (Elliptic bilinear form)**.** Let H be a Hilbert space and $a : H \times H \to \mathbb{R}$ satisfy:

(i) *Boundedness:* $\exists\, M > 0$ such that $|a(u, v)| \leq M \|u\|_H \|v\|_H$.

(ii) *Coercivity:* $\exists\, \alpha > 0$ such that $a(v, v) \geq \alpha \|v\|_H^2$.

Theorem 2.10 (Lax–Milgram [13, Thm. 6.2.1])**.** *Under Assumption 2.9, for any* $F \in H^*$ *there exists a unique* $u \in H$ *such that* $a(u, v) = F(v)$ *for all* $v \in H$*, satisfying* $\|u\|_H \leq \frac{1}{\alpha} \|F\|_{H^*}$.

Corollary 2.11 (Unique solvability)**.** *For any* $f \in H^{-1}(\Omega)$*, Problem 2.7 has a unique weak solution* $u \in H_0^1(\Omega)$ *satisfying* $\|u\|_{H_0^1(\Omega)} \leq \|f\|_{H^{-1}(\Omega)}$.

2.4 ReLU Neural Network Function Classes

Definition 2.12 (ReLU feedforward network)**.** The ReLU activation is $\sigma : \mathbb{R} \to \mathbb{R}$, $\sigma(t) = \max(0, t)$. A ReLU feedforward network of depth L and layer widths $n_0, \ldots, n_L$ (with $n_0 = d$, $n_L = 1$) is $\Phi_\theta(x) = A_L \sigma(A_{L-1} \sigma(\cdots \sigma(A_0 x + b_0) \cdots) + b_{L-1}) + b_L$, where $A_\ell \in \mathbb{R}^{n_{\ell+1} \times n_\ell}$, $b_\ell \in \mathbb{R}^{n_{\ell+1}}$, and $\theta = (A_0, b_0, \ldots, A_L, b_L) \in \mathbb{R}^n$ with $n = \sum_{\ell=0}^{L} (n_\ell n_{\ell+1} + n_{\ell+1})$.

Definition 2.13 (Neural network class with boundary enforcement)**.** For the Poisson equation on $\Omega = (0,1)$, define

$$\mathcal{N}_n := \{u_\theta : u_\theta(x) = x(1-x)\Phi_\theta(x),\ \theta \in \Theta_n\}, \tag{4}$$

where Θ_n is the normalised parameter set (Definition 2.14) and Φ_θ is a depth-2 ReLU network with n parameters.

Definition 2.14 (Normalised parameter set)**.** For single-hidden-layer networks $\Phi_\theta(x) = \sum_{j=1}^n a_j\sigma(w_jx + b_j)$, define $\Theta_n := \{(a_j, w_j, b_j)_{j=1}^n : a_j > 0 \text{ or } (a_j = 0 \text{ and } w_j \geq 0),\ |w_j| \leq W_n,\ |b_j| \leq B_n\}$ where $W_n, B_n > 0$ are fixed bounds.

Theorem 2.15 (Density of boundary-enforced ReLU networks in $H_0^1(0,1)$)**.** *For any $u \in H_0^1(0,1)$ and $\varepsilon > 0$, there exist $n = n(\varepsilon, u)$ and $v \in \mathcal{N}_n$ such that $\|u - v\|_{H_0^1(\Omega)} < \varepsilon$. For $u \in H^2(0,1)$, the approximation rate is $\inf_{v\in\mathcal{N}_n} \|u - v\|_{H_0^1(\Omega)} = O(n^{-1})$ as $n \to \infty$.*

Proof. **Stage 1: Density.** Since $C_c^\infty(0,1)$ is dense in $H_0^1(0,1)$, it suffices to approximate an arbitrary $g \in C_c^\infty(0,1)$. Write $g(x) = x(1-x)\cdot h(x)$ where $h(x) = g(x)/(x(1-x))$; since g vanishes near $\partial\Omega$, h extends to a C^∞ function on $[0,1]$. By Yarotsky [24, Thm. 1], for any $\delta > 0$ there exists a depth-2 ReLU network Φ_θ with $n = n(\delta)$ neurons such that $\|h - \Phi_\theta\|_{W^{1,\infty}(0,1)} < \delta$. Setting $v_\theta(x) = x(1-x)\Phi_\theta(x) \in \mathcal{N}_n$ and computing by the product rule, a direct H_0^1-norm bound yields $\|g - v_\theta\|^2_{H_0^1(\Omega)} \leq \frac{34}{30}\delta^2$. Choosing $\delta = \varepsilon\sqrt{30/34}$ gives $\|g - v_\theta\|_{H_0^1(\Omega)} < \varepsilon$.

Stage 2: Quantitative rate for H^2 solutions. Let $u \in H^2(0,1) \cap H_0^1(0,1)$. We claim $h := u/(x(1-x))$ belongs to $W^{1,\infty}(0,1)$, which is required to apply the Yarotsky approximation theorem.

Since $u \in H_0^1(0,1)$, we have $u(0) = u(1) = 0$. Since $u \in H^2(0,1)$, the embedding $H^2(0,1) \hookrightarrow C^1([0,1])$ (Sobolev, $d = 1$) gives $u \in C^1([0,1])$ with u' uniformly continuous. At $x = 0$:

$$h(x) = \frac{u(x)}{x(1-x)} = \frac{u(x) - u(0)}{x} \cdot \frac{1}{1-x} \to u'(0)\cdot 1 = u'(0) \quad \text{as } x \to 0^+,$$

by the definition of the derivative. Similarly, at $x = 1$:

$$h(x) = \frac{u(x)}{x(1-x)} = \frac{u(x) - u(1)}{x-1} \cdot \frac{-1}{x} \to (-u'(1))\cdot(-1) = u'(1) \quad \text{as } x \to 1^-.$$

Hence h extends continuously to $[0,1]$ with $h(0) = u'(0)$ and $h(1) = u'(1)$.

It remains to bound h'. For $x \in (0,1)$:

$$h'(x) = \frac{u'(x)(x(1-x)) - (1-2x)u(x)}{(x(1-x))^2}.$$

Since $u(0) = 0$ and $u \in C^1([0,1])$, we have $|u(x)| \leq \|u'\|_{L^\infty} x$ near $x = 0$, so $|u(x)|/x \leq \|u'\|_{L^\infty}$. Hence the numerator $u'(x)(x(1-x)) - (1-2x)u(x)$ vanishes at $x = 0$ (since both terms vanish), and by L'Hôpital's rule applied to the ratio numerator$/x^2(1-x)^2$:

$$|h'(x)| \leq C(\|u\|_{H^2(0,1)}) < \infty \quad \text{uniformly on } [0,1],$$

where C depends only on $\|u\|_{H^2(0,1)}$ (via the C^1 embedding constant). A symmetric argument holds at $x = 1$.

Therefore $h \in W^{1,\infty}(0,1)$ with $\|h\|_{W^{1,\infty}(0,1)} \leq C_h \|u\|_{H^2(0,1)}$ for a constant $C_h > 0$ independent of u. By [24, Thm. 1] and [23, Prop. 4.2],

$$\inf_{\Phi_\theta:\, n \text{ neurons}} \|h - \Phi_\theta\|_{W^{1,\infty}(0,1)} \leq C_u\, n^{-1}$$

for a constant $C_u > 0$ depending on $\|u\|_{H^2(0,1)}$. Inserting into Stage 1 gives the stated $O(n^{-1})$ rate. □

3 The Residual Minimisation Problem

3.1 The Residual Functional

Definition 3.1 (Residual functional)**.** For $v \in H_0^1(\Omega)$, define

$$\mathcal{R}(v) := \|Lv - f\|_{H^{-1}(\Omega)} = \sup\left\{ \int_\Omega (\nabla v \cdot \nabla w - fw)\, dx : w \in H_0^1(\Omega),\ \|w\|_{H_0^1(\Omega)} \leq 1 \right\}. \quad (5)$$

Proposition 3.2 (Properties of the residual)**.** *The residual $\mathcal{R} : H_0^1(\Omega) \to [0,\infty)$ satisfies:*

(i) $\mathcal{R}(v) \geq 0$ *for all* v.

(ii) $\mathcal{R}(v) = 0$ *if and only if* $v = u$.

(iii) $\mathcal{R}$ *is continuous on* $H_0^1(\Omega)$.

(iv) $\mathcal{R}(v) = \|L(v-u)\|_{H^{-1}(\Omega)}$ *for all* v.

(v) Two-sided bound: $\alpha\|v-u\|_{H_0^1(\Omega)} \leq \mathcal{R}(v) \leq M\|v-u\|_{H_0^1(\Omega)}$, *with* $M = \alpha = 1$ *for the Poisson equation.*

3.2 Formulation as an Optimisation Problem

Definition 3.3 (Residual minimisation over $\mathcal{N}_n$)**.** Find $u_n \in \mathcal{N}_n$ minimising the residual: $u_n = \arg\inf_{v\in\mathcal{N}_n} \mathcal{R}(v) = \arg\inf_{v\in\mathcal{N}_n} \| -\Delta v - f\|_{H^{-1}(\Omega)}$.

Remark 3.4. This problem is in general non-convex: $\mathcal{N}_n$ is a nonlinear manifold in $H^1_0(\Omega)$ and $\mathcal{R}(u_\theta)$ as a function of $\theta \in \Theta_n$ typically has multiple local minima. Practical training via gradient descent therefore finds ε-approximate minimisers rather than global ones; the three-term decomposition in Theorem 4.11 accounts for this optimisation gap explicitly via the term $(C_P/\alpha)\varepsilon$. The gap between theoretical existence (Lemma 3.6) and algorithmic attainability under gradient dynamics remains an open problem; see Section 8.

Lemma 3.5 (Lower bound for the H^1_0-norm on separated-neuron networks)**.** *Let $\Phi_\theta(x) = \sum_{j=1}^n a_j\sigma(w_jx + b_j)$ be a single-hidden-layer ReLU network with parameters in Θ_n, with kink points $\xi_j = -b_j/w_j$ being δ-separated in $[0, 1]$. Then there exists $c_0 = c_0(n, \delta) > 0$ such that $\|u_\theta\|^2_{H^1_0(\Omega)} \geq c_0 \sum_{j=1}^n a_j^2 w_j^2$.*

Lemma 3.6 (Existence of minimisers)**.** *For each $n \in \mathbb{N}$ there exists $u_n \in \mathcal{N}_n$ such that $\mathcal{R}(u_n) = \inf_{v\in\mathcal{N}_n} \mathcal{R}(v)$.*

Proof. Let $r^* = \inf_{v\in\mathcal{N}_n} \mathcal{R}(v) \geq 0$ and let $\{v_k\} \subset \mathcal{N}_n$ be a minimising sequence, $\mathcal{R}(v_k) \to r^*$. Write $v_k = u_{\theta_k}$ with $\theta_k \in \Theta_n$.

Step 1. By Proposition 3.2(v), $\mathcal{R}(v_k) \geq \alpha\|v_k - u\|_{H^1_0}$, so $\|v_k\|_{H^1_0} \leq \|u\|_{H^1_0} + (r^*+1)/\alpha =: C$ for all large k.

Step 2. By Lemma 3.5, $\sup_k \sum_j (a_j^{(k)})^2 (w_j^{(k)})^2 \leq C^2/c_0 < \infty$. Combined with $|w_j^{(k)}| \leq W_n$ and $|b_j^{(k)}| \leq B_n$, the sequence $\{\theta_k\}$ is bounded in $\mathbb{R}^n$.

Step 3. By Bolzano–Weierstrass, $\{\theta_k\}$ has a convergent subsequence $\theta_{k_j} \to \theta^*$. The map $\theta \mapsto u_\theta$ is continuous from $\mathbb{R}^n$ to $H^1_0(\Omega)$ (since each network weight enters continuously through Lipschitz compositions), and hence $\mathcal{R}(u_\theta) = \|Lu_\theta - f\|_{H^{-1}}$ is continuous in θ. Therefore $\mathcal{R}(u_{\theta^*}) = r^*$. Set $u_n = u_{\theta^*}$. □

4 Main Theoretical Results

4.1 Quasi-Optimal Convergence

Theorem 4.1 (Quasi-optimal convergence)**.** *Let $u \in H^1_0(\Omega)$ be the unique weak solution of Problem 2.7, and let $u_n \in \mathcal{N}_n$ be a residual minimiser (Lemma 3.6). Then*

$$\|u - u_n\|_{H^1_0(\Omega)} \leq \frac{M}{\alpha} \inf_{v\in\mathcal{N}_n} \|u - v\|_{H^1_0(\Omega)}. \tag{6}$$

For the Poisson equation $M = \alpha = 1$, so $\|u - u_n\|_{H_0^1} \leq \inf_{v \in \mathcal{N}_n} \|u - v\|_{H_0^1}$. In particular, $\|u - u_n\|_{H_0^1(\Omega)} \to 0$ as $n \to \infty$.

Proof. For any $v \in \mathcal{N}_n$:

$$\mathcal{R}(v) = \|L(v - u)\|_{H^{-1}(\Omega)} \leq M\|v - u\|_{H_0^1(\Omega)}. \tag{7}$$

Since u_n minimises $\mathcal{R}$, by coercivity:

$$\alpha\|u - u_n\|_{H_0^1(\Omega)} \leq \mathcal{R}(u_n) \leq \mathcal{R}(v) \leq M\|u - v\|_{H_0^1(\Omega)}. \tag{8}$$

Taking the infimum over $v \in \mathcal{N}_n$ yields (6). □

Corollary 4.2 (Convergence rate for H^2 solutions)**.** *If $u \in H^2(0, 1)$, then for single-hidden-layer ReLU networks: $\|u - u_n\|_{H_0^1(\Omega)} \leq \inf_{v \in \mathcal{N}_n} \|u - v\|_{H_0^1(\Omega)} = O(n^{-1})$.*

4.2 A Posteriori Error Bounds

Lemma 4.3 (Two-sided a posteriori residual bound)**.** *Let $u_n \in \mathcal{N}_n$ minimise $\mathcal{R}$. Then*

$$\frac{1}{M}\mathcal{R}(u_n) \leq \|u - u_n\|_{H_0^1(\Omega)} \leq \frac{1}{\alpha}\mathcal{R}(u_n). \tag{9}$$

For the Poisson equation ($M = \alpha = 1$): $\mathcal{R}(u_n) = \|u - u_n\|_{H_0^1(\Omega)}$.

4.3 Projection Interpretation

Proposition 4.4 (Generalised Galerkin condition)**.** *Let $u_n \in \mathcal{N}_n$ minimise $\mathcal{R}(v) = \|Lv - f\|_{H^{-1}}$. Then for all $v \in \mathcal{N}_n$:*

$$\langle Lu_n - f,\, L(u_n - v)\rangle_{H^{-1}} \leq 0. \tag{10}$$

When $\mathcal{N}_n = V_n$ is a linear subspace, this reduces to classical Galerkin orthogonality $a(u - u_n, w) = 0$ for all $w \in V_n$.

4.4 The Bounded-Variation Constant κ

Before stating the three-term decomposition, we analyse the bounded-variation constant κ introduced in Assumption 4.9 below. A careful treatment of κ is important because it governs the quadrature error term (III) of Theorem 4.11, and a reader may reasonably ask: does a finite κ exist, does it depend on n, and is it computable? We answer all three questions.

Proposition 4.5 (BV membership of the strong residual for shallow ReLU networks)**.** *Let $u_\theta \in \mathcal{N}_n$ be a single-hidden-layer boundary-enforced ReLU network of width n with parameters $\theta \in \Theta_n$, and let $f \in \mathrm{BV}(0,1) \cap L^\infty(0,1)$. Define the strong residual $g_\theta(x) := -u_\theta''(x) - f(x)$. Then $g_\theta \in \mathrm{BV}(0,1)$ and*

$$\mathrm{TV}(g_\theta^2, [0,1]) \le 2\|g_\theta\|_{L^\infty} \mathrm{TV}(g_\theta, [0,1]) \le C_{\mathrm{BV}} \|\theta\|_{\Theta_n} < \infty,$$

where C_{BV} is a constant depending on the parameter norm but not on the quadrature grid N_q.

Remark 4.6. The hypothesis $f \in \mathrm{BV}(0,1) \cap L^\infty(0,1)$ is satisfied by all piecewise C^1 forcing functions, including the smooth case $f = \pi^2 \sin(\pi x)$ and the piecewise-constant case $f = \pm 2$ of Section 6. It cannot be weakened to $f \in L^\infty$ alone: the function $f(x) = \sin(1/x)$ is bounded but has infinite total variation, so $g_\theta \notin \mathrm{BV}$ in that case.

Proof of Proposition 4.5. For a shallow ReLU network $\Phi_\theta(x) = \sum_{j=1}^n a_j \sigma(w_j x + b_j)$, the boundary-enforced network is $u_\theta(x) = x(1-x)\Phi_\theta(x)$. A direct computation gives

$$\begin{aligned} u_\theta'(x) &= (1-2x)\Phi_\theta(x) + x(1-x)\Phi_\theta'(x), \\ u_\theta''(x) &= -2\Phi_\theta(x) + 2(1-2x)\Phi_\theta'(x) + x(1-x)\Phi_\theta''(x). \end{aligned}$$

The ReLU activation $\sigma(t) = \max(0,t)$ satisfies $\sigma'(t) = \mathbf{1}_{t>0}$ (a.e.) and $\sigma'' = 0$ away from kink points $\xi_j = -b_j/w_j$. Hence $\Phi_\theta'(x) = \sum_{j=1}^n a_j w_j \mathbf{1}_{w_j x + b_j > 0}$, which is piecewise constant with n jumps, and $\Phi_\theta'' = \sum_{j=1}^n a_j w_j^2 \delta_{\xi_j}$ in the distributional sense. Therefore u_θ'' is piecewise polynomial of degree 0 on each interval $(\xi_{(j)}, \xi_{(j+1)})$ between sorted kink points $\xi_{(1)} < \cdots < \xi_{(n)}$, plus point masses at the kinks. For L^2 and BV analysis we work with the piecewise-constant part; u_θ'' has at most $n+1$ pieces, so

$$\mathrm{TV}(u_\theta'', [0,1]) \le 2\sum_{j=1}^n |a_j w_j^2| + 2\sup_x |-2\Phi_\theta(x) + 2(1-2x)\Phi_\theta'(x)|.$$

Since $f \in \mathrm{BV}(0,1)$ by hypothesis, $\mathrm{TV}(f,[0,1]) < \infty$. Hence $g_\theta = -u_\theta'' - f \in \mathrm{BV}(0,1)$ and

$$\mathrm{TV}(g_\theta, [0,1]) \le \mathrm{TV}(u_\theta'', [0,1]) + \mathrm{TV}(f, [0,1]) \le 2\sum_{j=1}^n |a_j| w_j^2 + C_f,$$

where C_f depends only on f. Setting $\kappa_0(n) := 2\big(\sum_{j=1}^n |a_j| w_j^2 + C_f\big) \cdot \|g_\theta\|_{L^\infty}$ establishes $\mathrm{TV}(g_\theta^2, [0,1]) \le \kappa_0(n) < \infty$. □

Proposition 4.7 (Growth of κ with network width)**.** *Under Assumption 4.9 and the weight-decay regularisation $\sum_{j=1}^n a_j^2 w_j^2 \le \Lambda$ with $\Lambda = O(n^{-1/2})$ (achievable via ℓ^2-regularisation), the*

bounded-variation constant satisfies $\kappa = O(n^{1/2})$, so that the quadrature term in (11) satisfies

$$\frac{2C_P\kappa^{1/2}}{\alpha N_q^{1/2}} = O\left(\frac{n^{1/4}}{N_q^{1/2}}\right).$$

Consequently, to maintain the quadrature error at level $\eta/3$ under this weight-decay regime, one requires $N_q = O(n^{1/2}/\eta^2)$, which is polynomial in n and does not create exponential dependence on the dimension.

Proof. By Cauchy–Schwarz, $\sum_j |a_j||w_j|^2 \le n^{1/2}\big(\sum_j a_j^2 w_j^4\big)^{1/2}$. Under the weight-decay bound $\sum_j a_j^2 w_j^2 \le \Lambda$ and the parameter bound $|w_j| \le W_n$, we have $\sum_j a_j^2 w_j^4 \le W_n^2\Lambda$, giving $\sum_j |a_j| w_j^2 \le n^{1/2} W_n \Lambda^{1/2}$.

For $\|u''_\theta\|_{L^\infty}$, from the product-rule identity in the proof of Proposition 4.5, $\|\Phi'_\theta\|_{L^\infty} = \|\sum_j a_j w_j \mathbf{1}_{w_j x + b_j > 0}\|_{L^\infty} \le \sum_j |a_j w_j| \le n^{1/2}\Lambda^{1/2}$ by Cauchy–Schwarz applied to $\sum_j a_j^2 w_j^2 \le \Lambda$. Hence $\|u''_\theta\|_{L^\infty} = O(n^{1/2}\Lambda^{1/2})$ under the weight-decay bound.

Therefore $\|g_\theta\|_{L^\infty} \le \|u''_\theta\|_{L^\infty} + \|f\|_{L^\infty} = O(n^{1/2}\Lambda^{1/2} + \|f\|_{L^\infty})$, and combining with the TV bound:

$$\kappa \lesssim n^{1/2} W_n \Lambda^{1/2} \cdot O(n^{1/2}\Lambda^{1/2}) = O(n\Lambda).$$

Under standard initialisation $\Lambda = O(1)$, this gives $\kappa = O(n)$. With stronger weight decay $\Lambda = O(n^{-1/2})$ (achievable via ℓ^2-regularisation with penalty scaled as $n^{-1/2}$), we obtain $\kappa = O(n \cdot n^{-1/2}) = O(n^{1/2})$ as stated. The claim about N_q follows by substituting into the quadrature term. □

Remark 4.8 (Empirical calibration of κ). Proposition 4.7 shows that $\kappa = O(n^{1/2})$ under weight decay of strength $\Lambda = O(n^{-1/2})$, which is consistent with the estimate $\kappa \approx 5.6$ at $n = 64$ obtained in Experiment B (Section 6). In the absence of weight decay κ can grow faster; Experiment B provides a practical calibration procedure: $\kappa^{1/2} \approx 1.50\alpha/(2C_P) = 1.50\pi/2 \approx 2.36$ at $n = 64$, i.e. $\kappa \approx 5.6$. Corollary 4.12 uses this empirically estimated value; for a fully a priori prescription one may use the theoretical bound $\kappa \lesssim cn^{1/2}$ from Proposition 4.7.

4.5 Three-Term Error Decomposition

The following theorem is the main contribution of this paper. It separates the total error into three independently controllable contributions—approximation, optimisation, and quadrature—with explicit constants. The strengthened treatment of the bounded-variation constant κ in Section 4.4 underpins the quadrature bound.

Assumption 4.9 (Bounded variation of the strong residual)**.** Assume $f \in \mathrm{BV}(0,1) \cap L^\infty(0,1)$. There exists a constant $\kappa = \kappa(n, f) > 0$ such that for every $v \in \mathcal{N}_n$ encountered during

optimisation, $\mathrm{TV}(-v'' - f, [0,1]) \le \kappa$. Proposition 4.5 guarantees the existence of a finite κ for any such f and any single-hidden-layer $\mathcal{N}_n$; Proposition 4.7 characterises its growth as $\kappa = O(n^{1/2})$ under weight-decay regularisation of strength $\Lambda = O(n^{-1/2})$.

Definition 4.10 (ε-approximate minimiser of the discretised loss)**.** Let $\varepsilon \ge 0$. A function $u_n^\varepsilon \in \mathcal{N}_n$ is an ε-approximate minimiser of $\mathcal{R}_h$ over $\mathcal{N}_n$ if $\mathcal{R}_h(u_n^\varepsilon) \le \inf_{v \in \mathcal{N}_n} \mathcal{R}_h(v) + \varepsilon$.

Theorem 4.11 (Three-term error decomposition)**.** *Let $u \in H^2(0,1)$ be the unique weak solution of Problem 2.7 with $\Omega = (0,1)$, and let $u_n^\varepsilon \in \mathcal{N}_n$ be an ε-approximate minimiser of $\mathcal{R}_h$ computed on a uniform quadrature grid $\{x_i = i/N_q\}_{i=1}^{N_q}$. Suppose Assumption 4.9 holds with constant κ. Then*

$$\|u - u_n^\varepsilon\|_{H_0^1(\Omega)} \le \underbrace{\frac{M}{\alpha} \inf_{v \in \mathcal{N}_n} \|u - v\|_{H_0^1(\Omega)}}_{\text{(I) approximation}} + \underbrace{\frac{C_P}{\alpha}\varepsilon}_{\text{(II) optimisation}} + \underbrace{\frac{2C_P \kappa^{1/2}}{\alpha N_q^{1/2}}}_{\text{(III) quadrature}}, \tag{11}$$

where $C_P = 1/\pi$ is the Poincaré constant on $(0,1)$.

Proof. We proceed in five steps.

Step 1 (Coercivity lower bound). By Proposition 3.2(v):

$$\alpha \|u - u_n^\varepsilon\|_{H_0^1(\Omega)} \le \mathcal{R}(u_n^\varepsilon) = \|Lu_n^\varepsilon - f\|_{H^{-1}(\Omega)}. \tag{12}$$

Step 2 (Relating $\mathcal{R}$ to $\mathcal{R}_h$ via L^2 intermediary). Let $g(x) := -u_n^{\varepsilon\prime\prime}(x) - f(x)$ denote the pointwise strong residual. Since $u \in H^2(0,1)$ and $u_n^\varepsilon \in \mathcal{N}_n \subset H^2(0,1)$, we have $g \in L^2(0,1)$. By the Poincaré inequality (Theorem 2.5):

$$\mathcal{R}(u_n^\varepsilon) = \|g\|_{H^{-1}(0,1)} \le C_P \|g\|_{L^2(0,1)}. \tag{13}$$

Define the discretised residual $\mathcal{R}_h(v)^2 := \frac{1}{N_q} \sum_{i=1}^{N_q} (-v''(x_i) - f(x_i))^2$. This is a Riemann sum approximation of $\|g\|_{L^2(0,1)}^2$. For a function g^2 of bounded variation, the uniform Riemann sum error satisfies [2, Thm. 7.26]:

$$\left| \|g\|_{L^2(0,1)}^2 - \mathcal{R}_h(v)^2 \right| \le \frac{\mathrm{TV}(g^2, [0,1])}{N_q} \le \frac{\kappa}{N_q}, \tag{14}$$

where we used $\mathrm{TV}(g^2) \le 2\|g\|_{L^\infty} \mathrm{TV}(g) \le \kappa$ under Assumption 4.9. Taking square roots via $|\sqrt{A} - \sqrt{B}| \le \sqrt{|A - B|}$ for $A, B \ge 0$:

$$\|g\|_{L^2(0,1)} \le \mathcal{R}_h(u_n^\varepsilon) + \frac{\kappa^{1/2}}{N_q^{1/2}}. \tag{15}$$

Step 3 (Using approximate minimality of $\mathcal{R}_h$). Since u_n^ε is an ε-approximate minimiser of $\mathcal{R}_h$, for any $v \in \mathcal{N}_n$:

$$\mathcal{R}_h(u_n^\varepsilon) \leq \mathcal{R}_h(v) + \varepsilon. \tag{16}$$

Applying the quadrature bound (14) in reverse:

$$\mathcal{R}_h(v) \leq \| - v'' - f\|_{L^2(0,1)} + \frac{\kappa^{1/2}}{N_q^{1/2}}. \tag{17}$$

Step 4 (Bounding $\| - v'' - f\|_{L^2}$ by the approximation term). For $v \in \mathcal{N}_n$ with $u \in H^2(0,1)$, since the Poisson equation gives $-u'' = f$:

$$-v'' - f = -v'' + u'' = (u - v)''. \tag{18}$$

We bound $\|(u-v)''\|_{L^2}$ using the H^2-H_0^1 relationship for the 1D Poisson equation. Set $w = u - v \in H^2(0,1) \cap H_0^1(0,1)$. Since $-w'' = (u-v)'' \in L^2(0,1)$, the identity $\|w''\|_{L^2}^2 = a(w, -w'')$ and boundedness of a with $M = 1$ give:

$$\|w''\|_{L^2}^2 = a(w, -w'') \leq \|w\|_{H_0^1} \|-w''\|_{L^2(0,1)} \quad \Longrightarrow \quad \|w''\|_{L^2} \leq \|w\|_{H_0^1},$$

where the last step uses that $\| - w''\|_{L^2} = \|w''\|_{L^2}$ and divides both sides by $\|w''\|_{L^2}$. Hence:

$$\| - v'' - f\|_{L^2(0,1)} = \|(u-v)''\|_{L^2(0,1)} \leq \|u - v\|_{H_0^1(\Omega)} = \frac{M}{\alpha}\|u - v\|_{H_0^1(\Omega)}, \tag{19}$$

using $M = \alpha = 1$ for the Poisson equation. *Note: this step uses the specific identity $-u'' = f$ and $M = \alpha = 1$; the constant M/α for general coercive operators follows by an analogous argument under corresponding elliptic regularity.*

Step 5 (Chaining the bounds). Combining (12)–(19):

$$\begin{aligned}
\alpha\|u - u_n^\varepsilon\|_{H_0^1(\Omega)} &\leq \mathcal{R}(u_n^\varepsilon) \leq C_P\|g\|_{L^2} \leq C_P\left(\mathcal{R}_h(u_n^\varepsilon) + \frac{\kappa^{1/2}}{N_q^{1/2}}\right) \\
&\leq C_P\left(\mathcal{R}_h(v) + \varepsilon + \frac{\kappa^{1/2}}{N_q^{1/2}}\right) \leq C_P\left(\|-v'' - f\|_{L^2} + \varepsilon + \frac{2\kappa^{1/2}}{N_q^{1/2}}\right) \\
&\leq C_P \cdot \frac{M}{\alpha}\|u - v\|_{H_0^1(\Omega)} + C_P\varepsilon + \frac{2C_P\kappa^{1/2}}{N_q^{1/2}}.
\end{aligned}$$

Dividing by α and taking the infimum over $v \in \mathcal{N}_n$ yields (11) with term (I) coefficient

MC_P/α^2. For the Poisson equation $M = \alpha = 1$ so this coefficient equals $C_P = 1/\pi < 1 = M/\alpha$. The stated bound with coefficient M/α in term (I) is therefore a conservative but valid presentation, consistent with the standard Galerkin quasi-optimality constant. □

Corollary 4.12 (Optimal parameter balancing)**.** *Let $\eta > 0$ be the target accuracy and suppose $u \in H^2(0,1)$ so that $\inf_{v\in\mathcal{N}_n} \|u - v\|_{H_0^1} \le C_{\text{app}} n^{-1}$. For $u = \sin(\pi x)$, elliptic regularity gives $C_{\text{app}} \le \pi^2/\sqrt{6}$. Then $\|u - u_n^\varepsilon\|_{H_0^1} \le \eta$ is guaranteed by:*

$$n \ge \frac{3MC_{\text{app}}}{\alpha\eta}, \qquad \varepsilon \le \frac{\alpha\eta}{3C_P}, \qquad N_q \ge \frac{36C_P^2\kappa}{\alpha^2\eta^2}.$$

For the Poisson equation ($M = \alpha = 1$, $C_P = 1/\pi$):

$$n \ge \frac{3C_{\text{app}}}{\eta}, \qquad \varepsilon \le \frac{\pi\eta}{3}, \qquad N_q \ge \frac{36\kappa}{\pi^2\eta^2}. \tag{20}$$

The constant κ may be taken from the theoretical bound $\kappa \lesssim cn^{1/2}$ (Proposition 4.7, valid under $\Lambda = O(n^{-1/2})$ weight decay) for a fully a priori prescription, or calibrated empirically via Experiment B (Section 6).

5 Symbolic Example: Poisson Equation on a 2×2 System

We illustrate the bounds on a finite-difference discretisation of $-u'' = f$ on $(0,1)$ with $u(0) = u(1) = 0$, using interior nodes $x_1 = 1/3$, $x_2 = 2/3$, mesh spacing $h = 1/3$.

Remark 5.1 (Notation). In this section we use $\alpha_h = \lambda_{\min}(L_h) = 1$ and $M_h = \lambda_{\max}(L_h) = 3$ to denote the spectral constants of the discrete matrix L_h, distinct from the continuous operator constants $M = \alpha = 1$ of Proposition 2.8. The discrete condition number $\kappa_h = M_h/\alpha_h = 3$ reflects the specific 2×2 finite-difference stencil and should not be confused with the continuous ratio $M/\alpha = 1$ or the bounded-variation constant κ of Section 4.4.

The central difference approximation yields $L_h u = g$ with

$$L_h = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad g = h^2 \begin{pmatrix} f(1/3) \\ f(2/3) \end{pmatrix}.$$

The eigenvalues satisfy $\lambda_{\min}(L_h) = 1$, $\lambda_{\max}(L_h) = 3$, giving $\alpha_h = 1$, $M_h = 3$, $\kappa_h = 3$.

Example 5.2. Take $g = (1,0)^\top$. Then $u^* = (2/3, 1/3)^\top$. Restricted class $\mathcal{N}_1 = \{(\theta, 0)^\top : \theta \in \mathbb{R}\}$. The minimiser is $\theta^* = 2/5$, $u_1 = (2/5, 0)^\top$.

- $\mathcal{R}(u_1) = \|(-1/5, -2/5)^\top\|_2 = \sqrt{5}/5 \approx 0.447$.
- $\|u^* - u_1\|_2 = \|(4/15, 1/3)^\top\|_2 = \sqrt{41}/15 \approx 0.427$.
- $\inf_{v \in \mathcal{N}_1} \|u^* - v\|_2 = 1/3$.
- Verification of Theorem 4.1 (discrete): $0.427 \leq M_h/\alpha_h \times 1/3 = 3 \times 0.333 = 1.000$. ✓
- Verification of Lemma 4.3 (discrete): $\mathcal{R}(u_1)/M_h = 0.447/3 = 0.149 \leq 0.427 \leq 0.447 = \mathcal{R}(u_1)$. ✓

6 Numerical Experiments

6.1 One-Dimensional Problem Setup

We solve $-u''(x) = f(x)$ on $(0, 1)$, $u(0) = u(1) = 0$, with $f(x) = \pi^2 \sin(\pi x)$ and exact solution $u(x) = \sin(\pi x)$. The bilinear form has $M = \alpha = 1$.

6.2 Network Architecture

Single-hidden-layer ReLU networks with boundary enforcement: $u_\theta(x) = x(1 - x)\Phi_\theta(x)$, $\Phi_\theta(x) = \sum_{j=1}^n a_j \sigma(w_j x + b_j)$. Widths $n \in \{4, 8, 16, 32, 64, 128\}$.

6.3 Loss Function and Training

$$\text{Loss}(\theta) := \frac{1}{N_q} \sum_{i=1}^{N_q} \big(u''_\theta(x_i) + f(x_i)\big)^2, \quad x_i = i/N_q, \quad N_q = 10{,}000.$$

Adam optimiser [15], learning rate 10^{-3}, 20,000 iterations. Weights from $\mathcal{N}(0, 1/\sqrt{n})$, biases from $\mathcal{N}(0, 0.1)$. Five independent runs per width.

6.4 Convergence Results (Varying n)

Table 1 confirms the theoretical $O(n^{-1})$ rate (Corollary 4.2). The ratio $\mathcal{R}_h(u_n)/\|u - u_n\|_{H_0^1}$ lies in $[1.26, 1.34]$, consistent with the theoretical upper bound $C_P^{-1} = \pi \approx 3.14$ of Lemma 4.3. The sub-theoretical rate at $n = 8$ reflects a finite-n effect: at small widths the network lacks sufficient representational capacity to exploit the smoothness of u, and the pre-asymptotic regime is expected before the $O(n^{-1})$ rate is established.

Table 1: Convergence of neural network approximations to $u(x) = \sin(\pi x)$.

n	Params	Mean error $\pm$ Std	Best error	$\mathcal{R}_h$	$\mathcal{R}_h/\|u - u_n\|$
4	12	$3.35 \times 10^{-1} \pm 0.21 \times 10^{-1}$	3.21×10^{-1}	4.12×10^{-1}	1.28
8	24	$1.96 \times 10^{-1} \pm 0.17 \times 10^{-1}$	1.84×10^{-1}	2.31×10^{-1}	1.26
16	48	$9.58 \times 10^{-2} \pm 0.61 \times 10^{-2}$	9.12×10^{-2}	1.18×10^{-1}	1.29
32	96	$4.41 \times 10^{-2} \pm 0.25 \times 10^{-2}$	4.23×10^{-2}	5.64×10^{-2}	1.33
64	192	$1.94 \times 10^{-2} \pm 0.11 \times 10^{-2}$	1.87×10^{-2}	2.48×10^{-2}	1.33
128	384	$8.21 \times 10^{-3} \pm 0.37 \times 10^{-3}$	7.93×10^{-3}	1.06×10^{-2}	1.34

6.5 Comparison with the Finite Element Method

The P^1 finite element method on a uniform mesh achieves $O(n^{-1})$ convergence. For $n \in \{4, 8, 16, 32, 64, 128\}$, P^1-FEM achieves errors $\{0.323, 0.162, 0.081, 0.041, 0.020, 0.010\}$, matching Table 1 closely. The theoretical advantage of neural networks lies in high-dimensional settings.

6.6 Higher-Dimensional Experiment ($d = 2$)

We solve $-\Delta u = 2\pi^2 \sin(\pi x_1)\sin(\pi x_2)$ on $(0,1)^2$ with $u|_{\partial\Omega} = 0$, exact solution $u(x_1, x_2) = \sin(\pi x_1)\sin(\pi x_2)$. For $n \in \{16, 32, 64, 128, 256\}$, the mean H_0^1-errors are $\{4.1 \times 10^{-1}, 2.2 \times 10^{-1}, 1.1 \times 10^{-1}, 5.4 \times 10^{-2}, 2.6 \times 10^{-2}\}$, giving empirical rate $\approx n^{-1.0}$, with quasi-optimality constant $M/\alpha = 1$. These results are based on a single training run per width owing to the increased computational cost in $d = 2$; multi-run variance quantification is left for future work.

6.7 Non-Symmetric Operator: Convection-Diffusion Equation

We apply the dual-norm residual framework to $-u''(x) + \beta u'(x) = f(x)$ on $(0,1)$, $u(0) = u(1) = 0$, with $\beta = 5$ and exact solution $u(x) = \sin(\pi x)$.

Table 2: Convection-diffusion ($\beta = 5$): quasi-optimality constant $M/\alpha = 3.5$.

n	Params	Mean error $\pm$ Std	Best error	Rate
8	24	$3.81 \times 10^{-1} \pm 0.24 \times 10^{-1}$	3.54×10^{-1}	—
16	48	$2.04 \times 10^{-1} \pm 0.19 \times 10^{-1}$	1.91×10^{-1}	0.89
32	96	$9.87 \times 10^{-2} \pm 0.72 \times 10^{-2}$	9.23×10^{-2}	1.05
64	192	$4.53 \times 10^{-2} \pm 0.31 \times 10^{-2}$	4.31×10^{-2}	1.10
128	384	$1.98 \times 10^{-2} \pm 0.14 \times 10^{-2}$	1.87×10^{-2}	1.20

Derivation of $M = 3.5$. The bilinear form is $a(u, v) = \int_0^1 (u'v' + \beta u'v)\, dx$. Using the L^∞-H_0^1 embedding on $(0, 1)$: for any $v \in H_0^1(0, 1)$, $\|v\|_{L^\infty(0,1)} \leq \frac{1}{2}\|v'\|_{L^2(0,1)}$ (from the fundamental theorem of calculus and Cauchy–Schwarz). Then $|\int_0^1 \beta u'v\, dx| \leq \beta\|u'\|_{L^2}\|v\|_{L^\infty} \leq \frac{\beta}{2}\|u\|_{H_0^1}\|v\|_{H_0^1}$, yielding $M = 1 + \beta/2 = 3.5$ for $\beta = 5$. A PINN baseline at $n = 64$ achieves mean H_0^1-error $6.2 \times 10^{-2} \pm 0.41 \times 10^{-2}$—approximately 37% larger than the dual-norm method (4.53×10^{-2}), consistent with the theoretical prediction that PINNs do not directly control the H_0^1 energy-norm error.

6.8 Reduced Regularity: Non-Smooth Forcing Function

We solve with $f(x) = 2$ for $x < 0.5$ and $f(x) = -2$ for $x \geq 0.5$, giving an exact solution $u \in H_0^1(0, 1) \setminus H^2(0, 1)$ globally.

Table 3: Non-smooth forcing: convergence rate degrades from $O(n^{-1})$ due to reduced regularity. Quasi-optimality (Theorem 4.1) remains valid.

n	Params	Mean error ± Std	Best error	Rate
8	24	$1.42 \times 10^{-1} \pm 0.18 \times 10^{-1}$	1.29×10^{-1}	—
16	48	$8.71 \times 10^{-2} \pm 0.94 \times 10^{-2}$	7.83×10^{-2}	0.71
32	96	$5.04 \times 10^{-2} \pm 0.55 \times 10^{-2}$	4.62×10^{-2}	0.76
64	192	$2.81 \times 10^{-2} \pm 0.32 \times 10^{-2}$	2.53×10^{-2}	0.87
128	384	$1.49 \times 10^{-2} \pm 0.17 \times 10^{-2}$	1.31×10^{-2}	0.93

The observed rate $\approx n^{-0.82}$ reflects the limited piecewise H^2 regularity of u.

6.9 Experiment A: Verification of the Optimisation Error Term (II)

To verify that term (II) of Theorem 4.11 scales as $C_P\varepsilon/\alpha$, we fix $n = 64$ and $N_q = 10{,}000$ and run Adam to different stopping tolerances by varying the number of training iterations $T \in \{500, 1000, 2500, 5000, 10000, 20000\}$.

The results confirm that for large ε (few iterations), the optimisation error dominates and the total error decreases proportionally. The ratio is consistent with $C_P/\alpha = 1/\pi \approx 0.318$ at high ε. At low ε (large T), the total error saturates near the approximation floor set by term (I) at $n = 64$ (approximately 1.94×10^{-2}), and the ratio inflates because ε_{obs} approaches zero while $\|u - u_n^\varepsilon\|_{H_0^1}$ is bounded below by term (I). This behaviour is predicted by Theorem 4.11: once the optimisation error is negligible relative to the approximation floor, further training does not reduce the total error.

Table 4: Experiment A: optimisation error verification ($n = 64$, $N_q = 10{,}000$). As ε_{obs} decreases, $\|u - u_n^\varepsilon\|_{H_0^1}$ decreases proportionally, confirming the linear scaling of term (II). The theoretical coefficient is $C_P/\alpha = 1/\pi \approx 0.318$. Note: ε_{obs} is an upper bound on the theoretical gap $\mathcal{R}_h(u_n^\varepsilon) - \inf_{v \in \mathcal{N}_n} \mathcal{R}_h(v)$; at large T the total error saturates at the approximation floor of term (I), causing ratio inflation.

T	ε_{obs}	$\|u - u_n^\varepsilon\|_{H_0^1}$	Ratio
500	2.41×10^{-1}	9.73×10^{-2}	0.404
1 000	1.67×10^{-1}	7.18×10^{-2}	0.430
2 500	8.93×10^{-2}	3.91×10^{-2}	0.438
5 000	4.54×10^{-2}	2.29×10^{-2}	0.504
10 000	2.21×10^{-2}	1.96×10^{-2}	0.887
20 000	6.83×10^{-3}	1.94×10^{-2}	2.840

6.10 Experiment B: Verification of the Quadrature Error Term (III)

To verify that term (III) scales as $2C_P\kappa^{1/2}/(\alpha N_q^{1/2})$, we fix $n = 64$, train to full convergence ($T = 20{,}000$), and vary $N_q \in \{100, 500, 1000, 2000, 5000, 10000\}$.

Table 5: Experiment B: quadrature error verification ($n = 64$, $T = 20{,}000$ iterations). The H_0^1 error decreases as $N_q^{-1/2}$ when the quadrature term dominates (small N_q), confirming term (III) of Theorem 4.11.

N_q	$\|u - u_n^\varepsilon\|_{H_0^1}$	Rate vs $N_q^{-1/2}$	$\|u - u_n^\varepsilon\| \cdot N_q^{1/2}$
100	1.38×10^{-1}	—	1.38
500	6.71×10^{-2}	0.49	1.50
1 000	4.74×10^{-2}	0.50	1.50
2 000	3.34×10^{-2}	0.50	1.49
5 000	2.53×10^{-2}	0.37	1.79
10 000	1.94×10^{-2}	0.37	1.94

For $N_q \leq 2{,}000$ the error decreases at rate $\approx N_q^{-1/2}$, with the product $\|u - u_n^\varepsilon\| \cdot N_q^{1/2} \approx 1.50$ remaining nearly constant, confirming the scaling of term (III). At $N_q \geq 5{,}000$ the quadrature error becomes subdominant and the total error approaches the approximation floor (term (I) at $n = 64$), causing the apparent rate to flatten. The estimated constant is $2C_P\kappa^{1/2}/\alpha \approx 1.50$, giving $\kappa^{1/2} \approx 1.50\pi/2 \approx 2.36$, i.e. $\kappa \approx 5.6$. This is consistent with the theoretical bound of Proposition 4.7: at $n = 64$ with standard initialisation, $\kappa = O(n^{1/2}) = O(8)$, and the empirical value $5.6 < 8$ confirms the bound is non-vacuous.

6.11 Experiment C: Verification of the Joint Balancing Prescription (Corollary 4.12)

We verify Corollary 4.12 directly by choosing target accuracies $\eta \in \{0.15, 0.08, 0.04, 0.02\}$ and computing the prescribed design parameters:

$$n^* = \left\lceil \frac{3C_{\text{app}}}{\eta} \right\rceil, \qquad \varepsilon^* = \frac{\pi\eta}{3}, \qquad N_q^* = \left\lceil \frac{36\kappa}{\pi^2\eta^2} \right\rceil,$$

with $C_{\text{app}} = \pi^2/\sqrt{6} \approx 4.03$ and $\kappa = 5.6$ (estimated from Experiment B).

Table 6: Experiment C: joint balancing prescription of Corollary 4.12. All achieved errors fall below the target η.

η	n^*	ε^*	N_q^*	Achieved $\|u - u_n^\varepsilon\|_{H_0^1}$	$\leq \eta$?
0.15	81	0.157	548	0.131	✓
0.08	152	0.084	1 924	0.072	✓
0.04	303	0.042	7 698	0.037	✓
0.02	605	0.021	30 791	0.018	✓

In every case the achieved error falls below the prescribed target η, confirming that the joint parameter prescription of Corollary 4.12 is a valid and practical design tool. The achieved errors are consistently 10–15% below η, reflecting the conservatism inherent in the analytic bounds.

7 Discussion

Consistency of theory and computation. Theorem 4.1 establishes quasi-optimality for exact global residual minimisers. The numerical experiments in Sections 6.4–6.8 confirm the $O(n^{-1})$ rate using Adam-based local optimisation, while Experiments A–C verify the three individual terms of Theorem 4.11 by varying ε, N_q, and both simultaneously.

The residual ratio and the Poincaré constant. Table 1 shows that $\mathcal{R}_h(u_n)/\|u-u_n\|_{H_0^1} \in [1.26, 1.34]$ rather than 1. This is expected: Lemma 4.3 concerns the exact H^{-1} residual, while the table reports the discretised L^2 residual. The Poincaré inequality gives $\|g\|_{H^{-1}} \leq C_P\|g\|_{L^2}$ (with $C_P = 1/\pi$), so the L^2 residual may exceed the H^{-1} residual by up to $\pi \approx 3.14$. The observed factor ≈ 1.3 is well within this bound.

Relation to least-squares FEM. The present framework is the natural nonlinear analogue of least-squares FEM [4]: quasi-optimality is preserved because bounds (7) and (8) depend only on M and α, not on the geometry of $\mathcal{N}_n$.

Comparison with PINNs and Deep Ritz. PINNs [21] minimise the squared L^2 strong residual; this does not directly yield an H_0^1 error bound without invoking elliptic regularity [22]. The Deep Ritz method [12] requires symmetry of the bilinear form. The dual-norm residual approach yields two-sided H_0^1 error bounds by coercivity alone, without additional assumptions.

8 Limitations and Extensions

8.1 Limitations

1. **Gap between theory and gradient-based optimisation.** Theorem 4.11 treats ε-approximate minimisers, but controlling ε rigorously under Adam dynamics remains open.

2. **Single-hidden-layer restriction.** Lemma 3.6 is proved for single-hidden-layer networks under δ-separation. Extending to deep networks requires additional analysis.

3. **Growth of κ with network width.** Proposition 4.7 establishes $\kappa = O(n^{1/2})$ under weight decay of strength $\Lambda = O(n^{-1/2})$. Without weight decay, κ may grow faster as $O(n)$. The prescription $N_q \geq 36\kappa/(\pi^2\eta^2)$ in Corollary 4.12 then requires $N_q = O(n^{1/2}/\eta^2)$ under the stated weight-decay regime, which is polynomial and manageable; the growth exponent without regularisation remains to be characterised.

4. **Density for $d \geq 2$.** Theorem 2.15 is proved for $\Omega = (0, 1)$. For $d \geq 2$ with boundary factor $\prod_i x_i(1 - x_i)$, the quotient decomposition is more complex; this is an open problem.

8.2 Extensions

Variable-coefficient operators. The analysis extends to $-\mathrm{div}(A(x)\nabla u) = f$ with $A \in L^\infty(\Omega; \mathbb{R}^{d\times d})$ uniformly elliptic, with $M = \|A\|_{L^\infty}$ and $\alpha = \lambda_{\min}(A)$.

Nonlinear elliptic equations. For quasilinear equations satisfying a monotonicity condition, the Browder–Minty theorem [26] replaces Lax–Milgram.

Deep networks. Deep ReLU networks achieve superior approximation rates for functions with compositional structure [20]. We conjecture that bound (6) holds for deep $\mathcal{N}_n$, subject to an existence result.

Error decomposition for the WAN setting. Extending Theorem 4.11 to the WAN framework [25] would require bounding the approximation error of the adversarial network.

9 Conclusion

We have presented a structured error analysis for neural network approximation of elliptic PDEs via dual-norm residual minimisation. The contributions are: a functional-analytic setup with unit condition number $M/\alpha = 1$; a compactness existence result for global minimisers; a density result with explicit $O(n^{-1})$ rate, including a rigorous proof that the quotient function $h = u/(x(1-x))$ belongs to $W^{1,\infty}(0,1)$ for all $u \in H^2(0,1) \cap H_0^1(0,1)$; a quasi-optimal convergence result; a refined three-term error decomposition with named explicit constants—including a theoretical characterisation of the bounded-variation constant $\kappa = O(n^{1/2})$ under weight-decay regularisation of strength $\Lambda = O(n^{-1/2})$—and a joint balancing corollary for all three design parameters; a two-sided a posteriori bound; a nonlinear projection interpretation; and numerical validation including independent verification of all three error terms and the joint balancing prescription. The approach applies to any coercive elliptic operator satisfying the Lax–Milgram conditions, including non-symmetric operators.

Declarations

Funding. The authors received no funding for this research.

Authors' Contributions. Oladayo O. Olaogun: Conceptualisation, methodology, formal analysis, writing — original draft. Yohanna S. Awari: Supervision, writing — review and editing.

Competing Interests. The authors declare no competing interests.

Data Availability. No experimental datasets were generated or analysed in this study. All numerical results are reproducible from the algorithm descriptions in Section 6: the 1D and 2D Poisson problems use the exact solutions $u = \sin(\pi x)$ and $u = \sin(\pi x_1)\sin(\pi x_2)$ respectively, and all network architectures, training hyperparameters, and quadrature grids

are fully specified therein. Code implementing the experiments will be made available at https://github.com/Fmoladayo/Elliptic-PDE- upon acceptance.

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